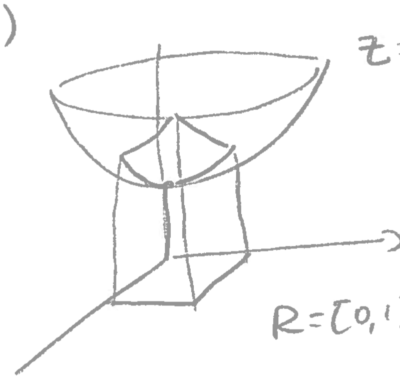
§ 13.2

$R = [a,b] \times [c,d]$

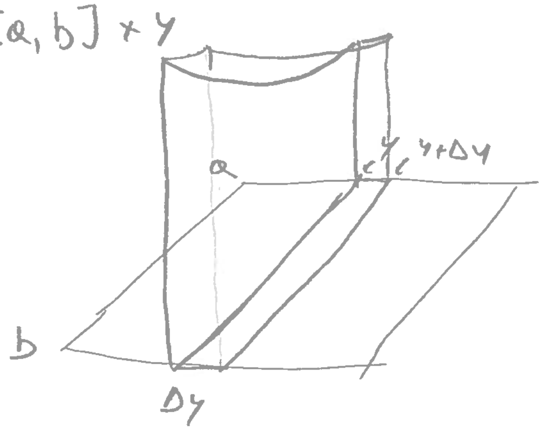
$$\iint_R f(x,y)\,dA = \int_c^d \left[\int_a^b f(x,y)\,dx\right] dy = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$$

<u>Explanation</u> $\iint_R f(x,y)\,dA$ represents the volume of the solid under $z = f(x,y)$



Fix y, then

$\int_a^b f(x,y)\,dx$ represents the area of the slice of this solid over the segment $[a,b] \times y$



$\left[\int_a^b f(x,y)\,dx\right] \times \Delta y$ is the volume of the slice of thickness Δy

The sum of all of these volumes converges to $\int_c^d \left[\int_a^b f(x,y)\,dx\right] dy$ as $\Delta y \to 0$.

Ex $$\iint_{[1,2]\times[1,2]} 2x - y\,dA = \int_1^2 \left(\int_1^2 (2x-y)\,dx\right) dy = \int_1^2 \left[x^2 - yx\right]_1^2 dy$$

$$= \int_1^2 \left(4 - 2y - (1-y)\right) dy = \int_1^2 (3-y)\,dy = \left[3y - \frac{y^2}{2}\right]_1^2 = 6 - \frac{4}{2} - \left(3 - \frac{1}{2}\right)$$

$$= 3/2$$

$$\int_1^2\int_1^2 (2x-y)\,dy\,dx = \int_1^2 \left[2xy-\frac{y^2}{2}\right]_1^2 dx = \int_1^2 4x-2-\left(2x-\frac{1}{2}\right)dx$$

$$= \int_1^2 \left(2x-\frac{3}{2}\right)dx = \left[x^2-\frac{3}{2}x\right]_1^2 = 4-3-\left(1-\frac{3}{2}\right) = \frac{3}{2}$$

(same answer!)

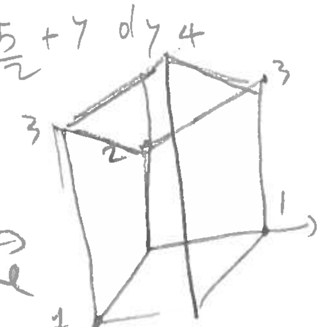
$$\iint_{[0,2]\times[0,2]} xy+2\,dA = \int_0^2\left(\int_0^2 (xy+2)\,dx\right)dy = \int_0^2\left[\frac{x^2y}{2}+2x\right]_0^2 dy$$

$$= \int_0^2 (2y+4)\,dy = \left[y^2+4y\right]_0^2 = 4+8 = 12$$

(Last time we proved $9 \le \iint_R xy+2\,dA \le 17$)

$$\iint_{[0,1]\times[0,1]} x+y+2\,dx\,dy = \int_0^1\left[\frac{x^2}{2}+xy+2x\right]_0^1 dy = \int_0^1 \frac{5}{2}+y\,dy$$

$$= \left[\frac{5}{2}y+\frac{y^2}{2}\right]_0^1 = \frac{5}{2}+\frac{1}{2} = 3 \xrightarrow{\text{volume}}$$



Area of base = 1
Average height = 3
Volume = 3

Ex1) $\int_0^2\int_1^3 x^2y\,dy\,dx \qquad \int_0^{\ln 3}\int_0^{\ln 2} e^{x+y}\,dy\,dx$

$\int_0^1\int_0^1 xe^{xy}\,dy\,dx \qquad \int_0^1\int_0^1 \frac{y}{(xy+1)^2}\,dx\,dy$

Ex2 Find the volume of the solid under $z = 1+x^2+y^2$ with $R=[0,4]\times[0,3]$